AI Powered Stock Market Insight Generator for the Colombo Stock Exchange

Project ID

R26-IT-076

Project Proposal Report

S A H Samarathunga – IT21296482

Component: Market Modeling and Explainable AI Insight Generation

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
S A H Samarathunga	**IT21296482**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

	Name	Date	Signature
Supervisor	Dr. Kapila Dissanayaka	20/03/2026.	
Co-Supervisor			

ABSTRACT

The rapid growth of digital trading platforms has increased the participation of retail investors in the Colombo Stock Exchange (CSE). However, many investors still struggle to interpret market behaviour using raw data such as historical prices, trading volumes, volatility levels, and technical indicators. As a result, investment decisions are often influenced by speculation, rumours, or incomplete analysis instead of evidence-based reasoning.

Although machine learning has been widely applied to stock market forecasting, many existing systems mainly focus on prediction accuracy and provide limited explanation for their outputs. This creates a major gap, especially in emerging markets like Sri Lanka, where investors need practical and understandable tools rather than complex numerical predictions alone.

To address this issue, this research proposes an explainable and anomaly-aware market modelling component within an AI-powered stock market insight system for the Colombo Stock Exchange. The proposed framework combines time-series analysis, regression-based pattern learning, and explainable AI to identify market trends, detect unusual stock behaviour, and explain the factors behind those changes.

The system further examines interactions among important market variables such as price, trading volume, and volatility in order to provide richer explanations for abnormal market conditions. In addition, a basic consistency check is introduced to assess whether the identified factor contributions remain stable over time, improving the reliability and trustworthiness of the generated insights.

The expected outcome of this research is a practical market analysis module that not only identifies patterns and anomalies, but also converts them into clear, investor-friendly explanations. This can support better financial literacy, more responsible investment decisions, and future business applications in financial analytics platforms and brokerage services.

Keywords: Explainable AI, Anomaly Detection, Stock Market Analysis, Time-Series Modeling, SHAP, Colombo Stock Exchange

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor and co-supervisor for their continuous guidance, valuable insights, and constructive feedback throughout the preparation of this research proposal. Their support and expertise played an important role in shaping the direction of this study and improving its overall quality. Their advice helped me gain clarity at different stages, especially while refining the research idea and methodology.

I would also like to thank the Department of Information Technology of the Sri Lanka Institute of Information Technology (SLIIT) for providing a supportive academic environment and the resources needed to carry out this research. The knowledge and experience I gained during my degree have been essential in completing this work.

In addition, I would like to thank my fellow research group members for their cooperation and the valuable discussions we had during the early stages of the project. Their ideas and feedback helped strengthen the overall concept of the proposed system and contributed to a more complete research approach.

I would also like to acknowledge the contributions of researchers and authors in the fields of artificial intelligence, machine learning, financial analytics, and explainable AI, whose work provided a strong academic foundation for this research.

Finally, I would like to express my heartfelt gratitude to my family and friends for their continuous encouragement, patience, and support throughout this journey. Their motivation played a significant role in helping me complete this proposal successfully.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
CSE	Colombo Stock Exchange
API	Application Programming Interface
CSV	Comma Separated Values
SHAP	SHapley Additive exPlanations
ARIMA	AutoRegressive Integrated Moving Average
LSTM	Long Short-Term Memory
RF	Random Forest
GBM	Gradient Boosting Machine
UI	User Interface
SDG	Sustainable Development Goals
DSR	Design Science Research

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INTRODUCTION

1.1 Background and Context

Stock markets play an important role in the financial system by helping companies raise capital while giving individuals and institutions opportunities to invest and build wealth. In Sri Lanka, the Colombo Stock Exchange (CSE) is the main platform for trading publicly listed company shares. Over the last decade, retail participation in the CSE has increased noticeably due to online trading platforms, wider internet access, and growing interest in alternative investment opportunities.

Despite this growth, many retail investors still find it difficult to analyze market information in a meaningful way. Understanding stock market behaviour requires more than just looking at prices. Investors often need to interpret trading volumes, volatility, technical indicators, company disclosures, and market sentiment. For people without a strong financial background, this information can be overwhelming.

As a result, many investment decisions are influenced by speculation, social media discussions, or market rumours instead of structured analysis. This creates a clear need for systems that can simplify complex market information and turn it into insights that ordinary investors can actually understand and use.

This issue affects multiple stakeholders. Retail investors may face avoidable financial losses due to limited analytical knowledge, while financial advisors and brokerage firms need better tools to present market information in a more accessible form. In emerging markets, these challenges are even more visible because investor behaviour is often driven by uncertainty, lower liquidity, and less mature financial literacy.

Existing platforms such as Bloomberg, Yahoo Finance, and TradingView provide valuable charts, indicators, and market data. However, these platforms are often designed for experienced users and are mostly centred on global markets. They do not fully address the needs of investors operating in local and emerging markets such as Sri Lanka. More importantly, they rarely explain market behaviour in simple language that helps investors understand why something is happening.

Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) have created new opportunities for financial analysis. These technologies can learn patterns from historical market data, identify hidden relationships, and support data-driven insights. However, many AI models still operate as black boxes. They may generate useful outputs, but they do not clearly show the reasoning behind those results. In financial decision-support contexts, that lack of transparency reduces trust.

In addition to transparency, another practical challenge is abnormal market behaviour. Sudden price spikes, unusual volume changes, or volatility surges can strongly affect investor decisions, yet many systems focus only on overall trend prediction and fail to explain such anomalies in a structured way.

To address these challenges, this research proposes an explainable and anomaly-aware market modelling framework as part of an AI-powered stock market insight platform designed for the Colombo Stock Exchange. The proposed component combines time-series analysis, regression-based pattern learning, and explainable AI to analyse historical market data, detect unusual behaviour, and explain the critical factors behind both normal and abnormal market conditions.

The framework also considers interactions among important variables such as price, trading volume, and volatility, because market behaviour is rarely driven by one factor in isolation. In addition, a basic consistency check is included to examine whether the identified factor contributions remain stable over time, thereby improving the reliability of the generated insights.

This component will operate within a larger system that also includes financial report analysis and sentiment analysis. The final output will be presented through a user-friendly dashboard that highlights trends, anomalies, risks, and explanations in a format that is easier for investors to understand. This research also aligns with SDG 8 and SDG 9 by promoting financial inclusion, responsible decision-making, and innovation through the use of advanced AI techniques in financial analysis.

1.2 Literature Review

Machine learning and artificial intelligence have become increasingly important in financial research, particularly in stock market forecasting and market behaviour analysis. Earlier studies relied heavily on statistical techniques such as AutoRegressive Integrated Moving Average (ARIMA) models. These methods were useful for modelling historical time-series patterns, but they often struggled to capture the nonlinear and dynamic nature of real market behaviour.

As research progressed, machine learning approaches such as Random Forest (RF) and Gradient Boosting Machine (GBM) gained attention for their ability to model more complex relationships in financial data. These methods can handle large feature sets, detect nonlinear interactions, and often deliver stronger performance than traditional statistical models when the market behaviour is not purely linear.

Deep learning methods, especially Long Short-Term Memory (LSTM) networks, further expanded the possibilities for financial forecasting. Because LSTM models are designed to learn sequential dependencies, they are well suited to time-series data where past behaviour influences future outcomes. In many studies, LSTM-based approaches have shown promising results for trend prediction and pattern recognition in stock markets.

At the same time, researchers have also begun using unstructured data such as financial news, company filings, and social media discussions. Natural Language Processing (NLP) techniques are often applied to extract sentiment and event-related signals, since investor mood and external events can have a direct impact on market movements.

Despite these advances, one major limitation remains: interpretability. Many machine learning and deep learning models generate outputs that are difficult for users to understand. In finance, accuracy alone is not enough. Investors and analysts also need to know why a model produced a certain result, especially when that result may influence a financial decision.

For this reason, explainable AI (XAI) has become an important area of research. Methods such as SHAP and LIME help reveal how input features contribute to model outputs. These techniques improve transparency by showing which variables have the strongest influence on predictions and how those influences change across cases.

More recent studies have also begun to explore explainability in stock market applications, but most of them still focus on explaining trend predictions rather than abnormal behaviour. In practice, anomalies such as sudden price jumps, unusual volume activity, or volatility shocks are often highly relevant to investors. However, structured and explainable anomaly detection remains underexplored in financial systems.

Another gap in the literature is the limited attention given to factor interaction and consistency over time. Market variables such as price, volume, and volatility often interact with one another, but many studies interpret them independently. Similarly, feature importance is often reported for one point in time without checking whether those contributions remain stable over different periods.

Industry tools such as AlphaSense, Kensho, and Bloomberg AI analytics offer advanced financial insights, but they are generally expensive and mainly designed for institutional or developed-market use. They are not specifically tailored for emerging markets such as Sri Lanka, where accessibility, local relevance, and investor education are especially important.

Overall, the literature shows strong progress in market prediction, pattern recognition, and model explainability. However, there is still a need for research that combines time-series modelling, anomaly detection, feature interaction analysis, and explainable AI in a way that is practical and understandable for investors in the Colombo Stock Exchange.

1.3 Research Gap

AI has made significant progress in stock market prediction, but several important gaps still remain. Most existing systems mainly focus on improving prediction accuracy without clearly explaining the factors behind their results. Investors are often presented with numerical outputs, but they are not given a clear understanding of what actually drives those predictions. This lack of transparency reduces trust and makes it difficult to use such systems confidently in real-world decision-making.

Another major limitation is that many AI-based financial platforms are developed for large markets such as the United States or Europe. In contrast, research focused on emerging markets like the Colombo Stock Exchange (CSE) is relatively limited. These markets have different characteristics, including unique investor behaviour, lower liquidity, and different economic influences. As a result, models developed for global markets may not perform effectively when applied to local contexts like Sri Lanka.

There is also a fragmentation issue in existing solutions. Most tools analyse market data, financial reports, and sentiment separately, rather than combining them into a single unified system. This separation limits the ability to generate deeper insights, as stock market behaviour is influenced by multiple interconnected factors.

Although explainable AI has gained attention in recent years, its practical application in financial systems is still limited. Many models continue to operate as black boxes, where users can see the results but not the reasoning behind them. This becomes a serious concern in financial environments where understanding the "why" behind a prediction is just as important as the prediction itself.

In addition, existing research rarely focuses on detecting abnormal or unusual market behaviour in a structured and explainable way. Most systems identify trends but fail to highlight anomalies or explain the reasons behind sudden market changes. Furthermore, the interaction between key market factors such as price, trading volume, and volatility is often overlooked. Another limitation is the lack of consistency evaluation, where systems do not verify whether identified patterns and feature contributions remain stable over time.

Therefore, there is a need for a more comprehensive approach that not only predicts market trends but also detects anomalies, explains their underlying causes, considers interactions between key factors, and ensures that insights remain reliable over time. This research addresses these gaps by integrating time-series modelling, regression-based pattern analysis, and explainable AI into a single framework tailored for the Colombo Stock Exchange, with the goal of generating clear and investor-friendly insights for both normal market behaviour and abnormal market conditions.

1.3.1 Research Gap Table

Function	Research Paper A	Research Paper B	Research Paper C	Proposed System
Historical stock market data analysis	✓	✓	✓	✓
Machine learning based market trend prediction	✓	✓	✓	✓
Integration of multiple data sources (market data, reports, news)	✗	✓	✗	✓
Explainable AI techniques to interpret predictions	✗	✗	✓	✓

Investor-friendly insight generation	✗	✗	✗	✓
Support for emerging markets such as CSE	✗	✗	✗	✓
Explainable anomaly detection	✗	✗	✗	✓
Factor interaction analysis	✗	✗	✗	✓
Consistency checking over time	✗	✗	✗	✓

OBJECTIVE

2.1 Main Objective

This research aims to develop an explainable and anomaly-aware market modelling framework that analyses historical data from the Colombo Stock Exchange and transforms it into clear, actionable insights for investors. The system not only identifies market trends and risks but also detects unusual market behaviour and explains the underlying reasons in a way that is easy to understand. The overall goal is to help investors gain a deeper understanding of market dynamics rather than relying solely on predictions.

2.2 Specific Objectives

To achieve this, the study focuses on the following objectives:

1. Develop a data collection and preprocessing pipeline to gather historical stock market data, including stock prices, trading volumes, and technical indicators from Colombo Stock Exchange sources.

2. Implement time-series and regression-based machine learning models to analyse market trends, identify patterns, and detect abnormal or unusual stock behaviour.

3. Apply Explainable Artificial Intelligence (XAI) techniques such as SHAP to interpret model outputs and identify the key factors influencing both normal market conditions and detected anomalies.

4. Analyse interactions between important market factors, such as price, trading volume, and volatility, to better understand how combined effects contribute to market behaviour and anomalies.

5. Introduce a consistency checking mechanism to evaluate whether the identified factor contributions remain stable over time, ensuring the reliability of generated insights.

6. Convert complex analytical results into clear, investor-friendly explanations that highlight trends, risks, anomalies, and their underlying causes.

7. Develop API-based integration to allow the market modelling component to communicate with other system modules, such as financial report analysis and sentiment analysis.

8. Evaluate the system using both machine learning performance metrics and usability testing to ensure that the insights are accurate, reliable, and easy for investors to understand.

These objectives collectively address the key research challenge of improving how investors interpret stock market data. They form the foundation for developing a more transparent, reliable, and practical market analysis component within the overall AI-powered stock market insight platform.

METHODOLOGY

3.1 Research Approach

Figure 3: Proposed Research Methodology Workflow

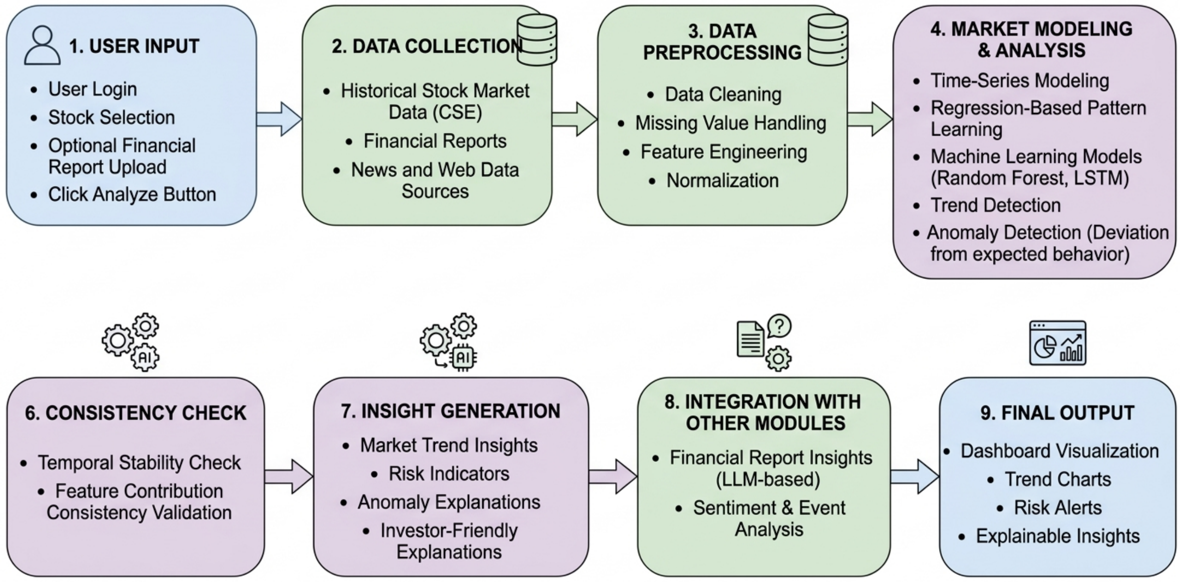

Figure 3

This study adopts a Design Science Research (DSR) methodology, because the main objective is not only to study the problem theoretically but also to design and build a practical solution. The artefact developed in this research is an explainable and anomaly-aware market modelling framework that analyses historical stock market data and generates investor-friendly insights.

DSR is suitable for this work because it supports a structured process of problem identification, artefact design, implementation, and evaluation. In this study, the process begins with identifying the limitations of current market analysis tools, followed by designing a system architecture, developing analytical models, integrating explainability techniques, and evaluating the usefulness of the generated insights.

The overall approach includes data collection, preprocessing, feature engineering, model development, anomaly detection, explainability analysis, consistency checking, insight generation, and user-oriented evaluation. Although this component is the main focus of the research, it is designed to operate within a larger modular platform that also includes financial report and sentiment analysis.

3.2 System Workflow

The system operates through a sequence of connected stages. First, the user logs in, selects a stock, uploads a financial report if available, and starts the analysis.

Once the process is triggered, the market modelling module retrieves historical market information such as price series, trading volume, and technical indicators. These data are then analysed using time-series and regression-based methods to identify trends, learn normal behaviour patterns, and detect unusual market movements.

At the same time, other system modules process financial reports and market sentiment from news or online content. The results from all modules are then combined into a unified dashboard, allowing investors to view market trends, anomalies, risk signals, and explanatory insights in one place.

3.3 Technical Methods and Algorithms

Figure 4: Data Processing and Machine Learning Pipeline for Stock Market Analysis

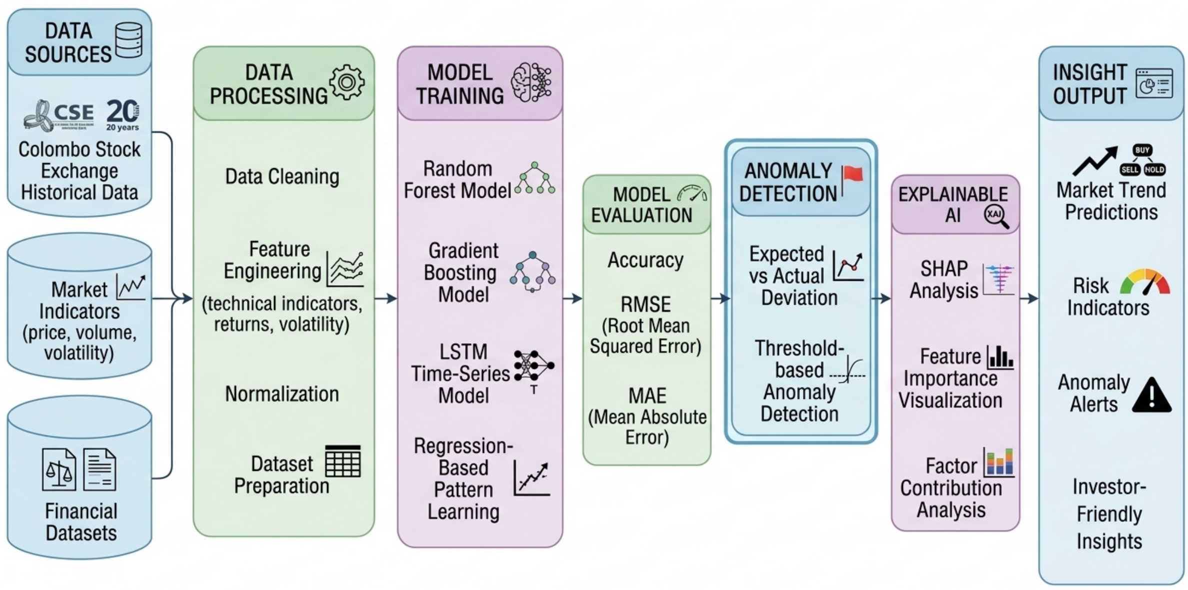

Figure 4

The proposed framework combines time-series analysis, regression-based pattern learning, and explainable AI techniques to analyse market behaviour in a more complete and practical way.

The process begins with preprocessing, including handling missing values, removing noise, normalising variables, and creating derived indicators such as moving averages, return-based measures, momentum, and volatility.

Next, time-series and machine learning models are used to learn market patterns. Models such as Random Forest, Gradient Boosting, and LSTM can be used to capture complex relationships and temporal dependencies. In addition, regression-based pattern learning is used to estimate expected behaviour and identify deviations between expected and observed values.

When these deviations become sufficiently large, they are treated as potential anomalies. In this way, the system does not only focus on predicting general trends, but also highlights unusual stock behaviour such as abnormal price movements, unusual volume patterns, or sudden volatility shifts.

To improve transparency, SHAP is applied to explain both predictions and anomaly-related behaviour. This helps identify which features contributed most to a given output and whether those contributions are positive, negative, or context-dependent.

The framework also examines interactions between important variables such as price, volume, and volatility. Instead of treating features in isolation, it looks at how combined effects influence market behaviour. A basic consistency check is then applied over time to assess whether the identified factor contributions remain stable across different periods.

Finally, the system converts technical outputs into clear explanations and concise visual insights, allowing investors to understand trends, anomalies, risk signals, and their likely causes without needing advanced technical knowledge.

3.4 Tools and Platforms

The system is implemented using widely used data science and software development tools. Python is used for data processing, model development, and integration tasks. Libraries such as Pandas and NumPy are used for data handling, while Scikit-learn supports machine learning workflows. TensorFlow or PyTorch may be used for deep learning models such as LSTM.

For time-series and regression-based analysis, tools from Scikit-learn and related Python libraries are used to model expected market behaviour and detect abnormal deviations. The SHAP library is used to generate explainability outputs, while backend integration can be handled through Python-based APIs or Node.js services depending on deployment needs.

Final insights are delivered through a web-based dashboard where users can interact with charts, anomaly markers, trend summaries, risk indicators, and plain-language explanations within a single interface.

3.5 Data Collection and Ethical Considerations

This study uses historical stock market data from the Colombo Stock Exchange, including stock prices, trading volumes, technical indicators, and related financial variables. Data will be obtained from publicly available or authorised sources. Since the research focuses on public market information, it does not involve personal or sensitive user data in the core analytical process.

Even so, ethical considerations remain important. The system is designed as a decision-support tool and not as a financial advisory system. It is intended to help investors better understand market behaviour, anomalies, and potential risks, rather than provide guaranteed recommendations or trading instructions.

All datasets and generated outputs will be handled responsibly, with attention to transparency, proper storage practices, and clear communication of system limitations.

3.6 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed framework, this study will use a combination of technical performance measures and usability-oriented assessment.

For market trend modelling, evaluation will include standard predictive metrics such as Accuracy, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), depending on the specific modelling task. For anomaly detection, performance can also be assessed using measures such as precision, recall, or F1-score where labelled or proxy anomaly conditions are available.

Interpretability will be evaluated by examining whether SHAP explanations meaningfully identify the factors influencing predictions and detected anomalies. In addition, the consistency mechanism will be reviewed to determine whether key factor contributions remain reasonably stable over time.

Usability evaluation will focus on whether the generated explanations, anomaly indicators, and visual outputs are understandable and useful to investors. This may include qualitative feedback and simple user-based assessment of clarity and perceived trustworthiness.

Through this combined evaluation strategy, the system will be assessed not only for technical performance, but also for its ability to produce clear, reliable, and investor-friendly insights.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The proposed solution is an AI-driven Stock Market Insight Generator designed for the Colombo Stock Exchange (CSE). Its purpose is not to tell investors what to buy or sell, but to help them interpret the market through clear, evidence-based insights. The system combines historical market data, financial report understanding, sentiment analysis, and explainable AI to provide a broader view of stock behaviour.

The architecture consists of several connected modules, each responsible for a different type of financial information. The process begins when an investor signs in, selects a stock, optionally uploads a financial report, and starts the analysis. The system then activates multiple analytical components that work in parallel and send their results to a shared integration layer.

The system operates as follows:

1. User Interface Layer

This layer allows investors to select stocks, upload reports, explore dashboards, and review generated insights.

2. Data Source Layer

This layer provides the core input data, including historical CSE market data, company financial statements, and relevant financial news or online sources.

3. Explainable and Anomaly-Aware Market Modelling Component

This core component analyses historical prices, volumes, volatility, and technical indicators using time-series analysis, regression-based pattern learning, and machine learning. It identifies trends, detects abnormal behaviour, explains feature contributions, examines factor interactions, and checks whether explanations remain consistent over time.

4. Financial Report Understanding Component

This module uses large language models to process uploaded annual or earnings reports and extract investor-relevant financial insights.

5. Sentiment and Event Analysis Component

This module analyses financial news and online sources using search and natural language processing to detect sentiment shifts and important market events.

6. Insight Integration Layer

This layer combines outputs from all modules and creates a unified summary of trends, anomalies, risks, business performance, and contextual market signals.

7. Output and Visualization Layer

Final insights are presented through an interactive dashboard with charts, anomaly indicators, explanatory panels, and risk summaries.

This modular structure allows each component to process its own data independently while still contributing to one coherent investor-facing insight.

4.2 System Diagram of Components

Figure 1: Overall System Architecture of AI Powered Stock Market Insight Generator for the Colombo Stock Exchange

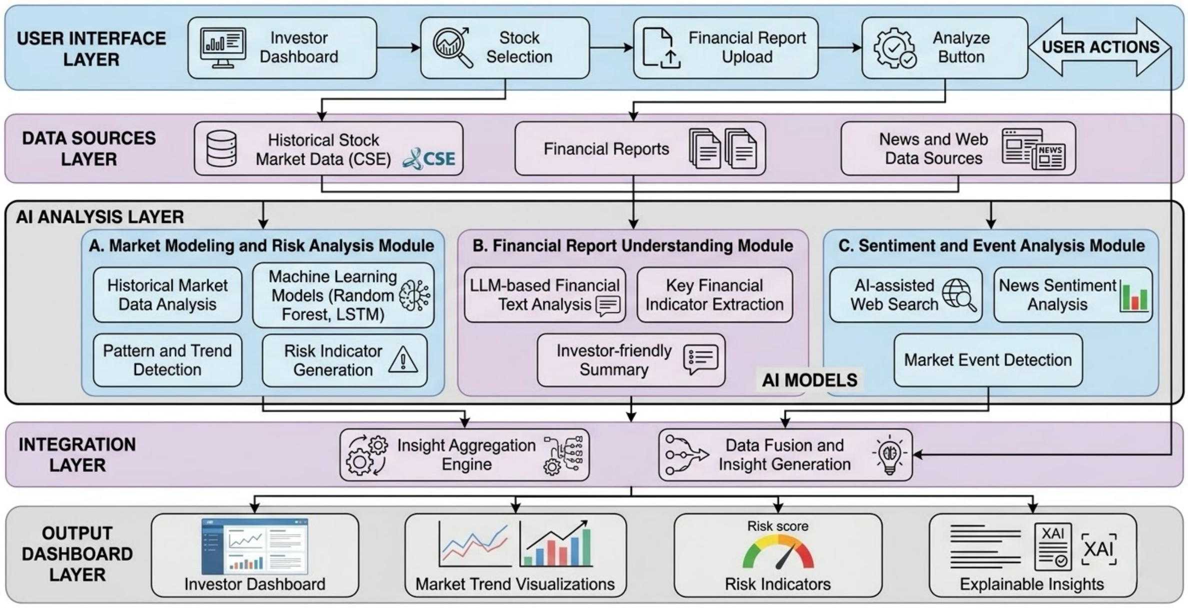

Figure 1

The system architecture can be understood as a layered workflow that starts with user input and ends with a unified insight dashboard.

User Interface → Data Sources → AI Analysis Components → Insight Integration Layer → Output Dashboard

There are four main analytical streams within the solution:

- analysis of historical market data
- interpretation of financial reports
- detection of sentiment and market events
- synthesis of integrated insights

The central research focus is the explainable and anomaly-aware market modelling component, where historical market data are transformed into trend signals, anomaly indicators, and transparent explanations for investors.

4.3 Individual Component and Its Internal Parts

Figure 2: Market Modeling and Explainable AI Component Architecture

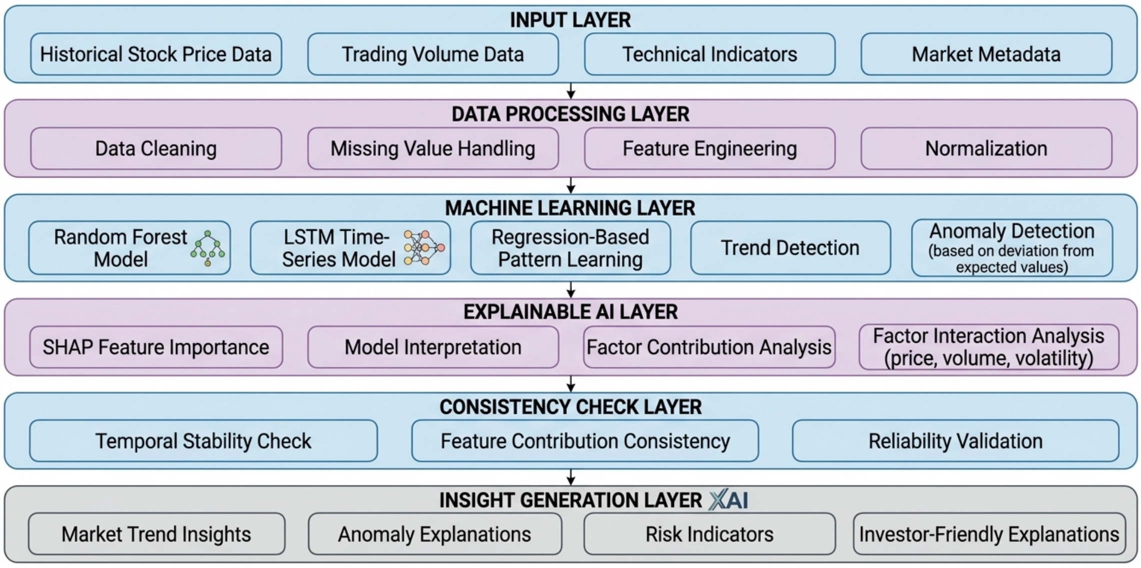

Figure 2

The segment assigned to this research is the Explainable and Anomaly-Aware Market Modelling and Insight Generation module, attributed to S A H Samarathunga (IT21296482).

Within this component:

4.3.1 Historical Market Data Acquisition

This module gathers historical CSE data such as opening and closing prices, highs and lows, trading volumes, and time-series records. These data form the basis for model training and market behaviour analysis.

4.3.2 Data Preprocessing and Feature Engineering

At this stage, raw market data are cleaned and prepared for modelling. Missing values are handled, outliers are reviewed, variables are normalised where necessary, and technical indicators such as moving averages, momentum, returns, and volatility are generated.

4.3.3 Time-Series and Machine Learning Modelling

This part of the component uses time-series methods, regression-based pattern learning, and machine learning models to analyse trends, estimate expected behaviour, and identify unusual market movements or risk conditions.

4.3.4 Explainable AI and Interaction Analysis

After the model produces outputs, explainability techniques are applied to identify which features influenced the result. SHAP is used to show how variables such as price changes, trading volume, and volatility contribute to both general market patterns and detected anomalies. The system also examines how these variables interact rather than considering them one by one.

4.3.5 Consistency Check and Insight Generation

The identified factor contributions are then reviewed over time to assess whether they remain stable across different periods. The final stage converts model outputs into clear, practical insights, including trend explanations, anomaly descriptions, risk indications, and the main reasons behind recent market behaviour.

4.3.6 API and Component Communication

These outputs are passed to the integration layer through APIs, where they are combined with insights from financial report analysis and sentiment analysis.

4.4 Component Interactions

The proposed solution includes several specialised modules that work independently but exchange their findings. When the user selects a stock and starts an analysis, the market modelling module retrieves historical data from the selected data source. If a report is uploaded, it is processed by the document understanding module. At the same time, relevant market news and online information are analysed by the sentiment and event module.

These modules contribute different kinds of intelligence:

the market modelling module provides trend signals, anomaly detection, and explainable risk insights

the financial report module evaluates company performance and report-based indicators

the sentiment module adds context from news, events, and public market mood

All outputs are sent to the Insight Integration Layer, where they are combined into a more complete and realistic view of market behaviour. This is important because stock movement is rarely driven by one data source alone. By combining quantitative market patterns with financial fundamentals and sentiment signals, the system produces richer and more useful investor insights.

Finally, the dashboard presents the integrated outcome in a form that is easy to review and understand.

4.5 Data Flow Explanation

Here is how data move through the system step by step:

Step 1: User Input

The investor logs in, chooses a stock, uploads a financial report if available, and clicks Analyze.

Step 2: Data Retrieval

The system collects historical market data, the uploaded company report, and relevant financial news or web information.

Step 3: Component-Level Processing

Each data type is routed to the appropriate module: market data to the modelling component, financial documents to the LLM-based component, and news or web data to the sentiment and event module.

Step 4: Internal Processing

Within the market modelling component, data are cleaned, features are engineered, normal behaviour patterns are learned, anomalies are detected, and explainability analysis is applied.

Step 5: Insight Generation

Each module produces its own outputs. The market component generates trend, anomaly, and factor-based explanations; the report module produces financial summaries; and the sentiment module identifies event-driven signals.

Step 6: Insight Fusion

The integration layer combines all outputs into a single investor-friendly summary.

Step 7: Dashboard Presentation

The dashboard displays trend charts, anomaly markers, risk indicators, explanation panels, and summary cards for quick interpretation.

This flow keeps the system organised, scalable, and easy to understand from a user perspective.

4.6 Scalability and Security Considerations

This architecture gives strong importance to both scalability and security.

Scalability

The modular design allows each component to be improved or expanded without disrupting the rest of the system. For example, the market modelling module can later include stronger anomaly detection methods, richer interaction analysis, or additional models without requiring a complete redesign. Likewise, the report and sentiment modules can be extended with new data sources when needed.

Because each module can operate as its own service or logical unit, the system can be deployed on cloud infrastructure and scaled according to workload. If market modelling or explainability tasks require more computational resources, those modules can be scaled independently.

The architecture is also suitable for future growth, including support for more listed stocks, longer historical datasets, additional technical indicators, stronger anomaly detection rules, and enhanced dashboard features.

Security

Although the system mainly uses public financial data, it still handles user accounts, uploaded documents, and system interaction data. For that reason, essential safeguards are required, including authenticated access control, secure API communication between modules, proper validation of uploaded files, controlled storage access, and protection against malicious or unauthorised use.

Importantly, the platform is designed as an insight support system rather than an automated financial advisor. This helps reduce ethical and legal concerns while keeping the focus on transparency, education, and informed decision support.

USER REQUIREMENTS

5.1 Stakeholder Identification

Identifying stakeholders at an early stage helps ensure that the platform is practical and relevant to everyone involved.

Retail Investors

Retail investors are the primary users of the platform. Many of them are not financial professionals and need support in understanding trends, risks, unusual market movements, and company-related information through clear explanations and simple visualisations.

Financial Analysts

Analysts can use the system to review historical market behaviour more efficiently, detect unusual signals, and understand which factors are driving stock changes through explainable outputs.

Brokerage Firms

Brokerage firms may use the platform to strengthen the analytics services they offer to clients and to improve client engagement with transparent, AI-assisted market insights.

Researchers and Developers

Researchers and developers in AI or fintech can use the system as a practical environment for testing market models, anomaly detection methods, and explainability techniques.

System Administrators

System administrators are responsible for maintaining the technical environment, monitoring performance, managing access, and ensuring smooth system operation.

5.2 Functional Requirements

These are the essential tasks the system must perform to deliver value:

FR1: Enable users to register accounts and log in securely.

FR2: Allow users to select any stock listed on the Colombo Stock Exchange for analysis.

FR3: Allow users to upload annual reports, earnings reports, or related company documents for system analysis.

FR4: Retrieve historical market data, including stock prices, trading volumes, and related indicators, for the selected company.

FR5: Apply time-series and machine learning methods to analyse market trends and behaviour patterns.

FR6: Detect unusual or abnormal market behaviour using regression-based pattern learning and anomaly-aware analysis.

FR7: Use explainable AI to identify the key factors influencing predictions and detected anomalies.

FR8: Analyse interactions between major market factors such as price, volume, and volatility.

FR9: Perform a consistency check to determine whether important factor contributions remain stable over time.

FR10: Analyse uploaded financial reports using large language models to generate investor-friendly summaries.

FR11: Collect relevant financial news and web data for sentiment analysis and event detection.

FR12: Integrate outputs from market modelling, report analysis, and sentiment analysis into a unified result.

FR13: Present trends, anomalies, risks, and explanations through an interactive dashboard.

FR14: Allow users to review previous analyses and past insight summaries.

5.3 Non-Functional Requirements

These requirements describe how the system should operate, not just what it should do.

Security

User accounts, uploaded files, and internal communications must be protected through authentication, secure API communication, file validation, and controlled data access.

Performance

The system should provide analysis results within a reasonable time even when handling large historical datasets or multiple analytical modules.

Usability

The interface should be easy to navigate for non-expert users, and the explanations should be written in a form that is clear and practical.

Reliability

The platform should remain stable even when external data sources are slow, incomplete, or temporarily unavailable. Error handling should minimise disruption.

Scalability

The architecture should support future expansion, including more stocks, more data sources, stronger models, and richer dashboard features.

Maintainability

The codebase should remain modular and easy to update so that models, anomaly rules, feature pipelines, or APIs can be improved without major redesign.

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary audience for the proposed system is retail investors in the Colombo Stock Exchange. In recent years, more individual investors have entered the market through online trading platforms, yet many still struggle to interpret technical indicators, company reports, or unusual market movements with confidence.

This platform mainly targets new and intermediate investors who want understandable, trustworthy analysis before making investment decisions. In addition, the system is relevant to financial analysts, brokerage firms, and academic users who need a practical way to explore transparent market insights.

The typical user is someone who actively follows the CSE through digital platforms but does not necessarily have deep expertise in technical analysis. What they need is not just raw numbers, but simple explanations of trends, anomalies, risks, and the factors influencing recent market behaviour.

6.2 Value Proposition

The core value of the platform is its ability to provide Sri Lankan investors with clear, data-driven market insights that are easier to understand and trust. Instead of presenting only forecasts or technical indicators, the system explains what is happening in the market and why it may be happening.

Behind the scenes, the platform combines market modelling, anomaly detection, financial document analysis, and sentiment analysis. This allows users to see not only expected market behaviour, but also unusual movements and the factors likely driving those changes.

A major differentiator is the use of explainable AI together with anomaly-aware analysis. Users are not limited to a prediction output; they also receive explanations of feature contributions, interactions among market variables, and a basic indication of whether those explanations remain consistent over time.

Another strong advantage is the platform’s focus on the Colombo Stock Exchange. By being tailored to an emerging local market, it offers more relevant and accessible insight than many global tools that are designed mainly for developed-market users.

6.3 Revenue Model

The business model is based on Software-as-a-Service (SaaS), where users access the platform online and pay for advanced analytical features.

A freemium approach can be used, where general trend views and limited insights are available for free, while premium subscriptions unlock deeper anomaly analysis, explainable AI insights, financial report interpretation, and richer dashboard features.

Brokerage firms, financial service providers, and educational institutions may also license the platform for client use, internal analysis, or training purposes. In addition, API access can create future revenue opportunities by allowing other financial platforms to integrate the platform’s analytics.

6.4 Cost Estimation

The system is built using open-source technologies, which helps keep development costs low. The main allocation of the budget focuses on essential areas: system development, cloud infrastructure, data collection, and ongoing maintenance.

Since university students are responsible for much of this work as part of an academic project, labor expenses are kept to a minimum.

Cloud infrastructure may introduce additional expenses, particularly for hosting the platform and managing servers that process financial data and execute the machine learning models. Storage requirements involve maintaining historical market data as well as any uploaded financial documents.

Operating the system also brings recurring costs: ensuring data pipelines remain functional, updating models, monitoring performance, and maintaining security.

Thanks to the use of public financial datasets and open-source tools, both development and operational costs are kept at a manageable level.

6.5 Pricing Strategy

The pricing approach must remain affordable for retail investors while also sustaining the platform over the long term. A tiered subscription model is well suited for this purpose.

The basic plan provides free access to general market insights and limited analysis. Users seeking more in-depth features can upgrade to a premium plan, which unlocks advanced analytics, explainable AI, financial report interpretation, and additional visualization options.

Enterprise packages are available for institutional clients such as brokerage firms or analytics providers, with pricing based on system usage, user counts, or API access.

Pricing should remain competitive with international financial analytics platforms, but must also be realistic for local Sri Lankan investors.

6.6 Competitive Advantage

This system offers several important competitive advantages.

Many existing platforms such as TradingView and Yahoo Finance focus heavily on charting and technical indicators. While useful, they often expect users to already understand how to interpret those signals.

Many AI-based tools improve prediction accuracy but still leave users with little visibility into the reasons behind the result. In contrast, this system combines prediction support, anomaly detection, explainable AI, factor interaction analysis, and consistency checking in one investor-oriented workflow.

Another advantage is its focus on the Colombo Stock Exchange, which remains underrepresented in most global financial analytics platforms. The system is therefore better aligned with local investor needs, market behaviour, and context.

Finally, by integrating historical market data, financial reports, and sentiment or event analysis into a single platform, the system offers a broader and more practical view of market behaviour than tools that depend on only one source of information.

6.7 Intellectual Property Considerations

The system introduces several innovative elements, including anomaly-aware market modelling, explainable insight generation, factor interaction analysis, consistency checking over time, and integration with financial document understanding and sentiment analysis.

Although many underlying libraries and algorithms are based on open-source technologies, the overall architecture, system workflow, data pipelines, dashboard logic, and the way these components are integrated can be treated as intellectual property.

Copyright protection may apply to the source code, interface design, visualisations, and system documentation. If the project develops into a commercial product, additional legal protection such as licensing arrangements or proprietary implementation strategies may also be considered.

At the same time, any external market data used by the system must be handled in accordance with data usage policies, licensing conditions, and ethical standards.

BUDGET AND JUSTIFICATION

Table 2: Budget Estimation

Category	Item	Description	Estimated Cost (LKR)
Hardware	Development Computer Resources	Personal computers used for development, testing, and running machine learning models	20,000
Hardware	Storage Devices	External storage devices for dataset storage and backup	10,000
Software	Development Tools	Integrated development environments, libraries, and supporting tools	5,000
Cloud Services	Cloud Computing Resources	Cloud servers used for machine learning model training and system deployment	30,000
Cloud Services	Data Storage	Cloud storage for financial datasets and system outputs	10,000
Human Effort	Development and Testing	Time spent by developers on system design, implementation, and testing	20,000
Miscellaneous	Documentation and Maintenance	Report preparation, documentation, and system maintenance costs	5,000

Total Estimated Budget = 100,000(LKR)

7.1 Hardware Costs

Developing and testing the proposed framework requires a reasonable level of computing power. Personal computers will be used for coding, model training, explainability analysis, anomaly testing, and general system development. External storage can be used to maintain datasets, backups, and generated outputs.

The main development tools, including Python, machine learning libraries, and web frameworks, are open source, which helps keep software costs low. However, some additional tools, plugins, or specialised utilities may still require small licensing or subscription costs depending on the development environment.

Cloud services are needed to support model training, deployment, storage, and integration across modules. These resources are especially useful when handling larger historical datasets, anomaly analysis workflows, and web-based access to the system.

Human effort is also an important part of the budget. Considerable time is required for data preparation, model building, explainability analysis, anomaly detection logic, API integration, dashboard development, testing, and documentation.

Overall, the budget is justified because it covers the essential resources needed to design, implement, test, and present the proposed system while still remaining cost-effective through the use of open-source technologies.

WORK BREAKDOWN STRUCTURE (WBS)

Table 3: Work Breakdown Structure

WBS ID	Project Phase	Task Description
1.0	Project Planning	Define project scope, objectives, and research problem
1.1	Literature Review	Review existing research on stock prediction, anomaly detection, machine learning, and explainable AI
1.2	Research Proposal Preparation	Prepare research proposal, documentation, and system design plan
2.0	Data Collection	Collect historical stock market data from the Colombo Stock Exchange
2.1	Financial Data Acquisition	Gather financial reports, earnings statements, and related documents for integrated analysis
2.2	News and Event Data Collection	Collect financial news and market event information for sentiment and context analysis
3.0	Data Processing	Perform data cleaning, preprocessing, and feature engineering for market modelling
3.1	Feature Extraction	Generate technical indicators, volatility measures, and regression-based behaviour features
4.0	Model Development	Develop time-series and machine learning models for market trend and anomaly analysis
4.1	Explainable AI Implementation	Apply SHAP and interaction analysis to interpret model outputs and detected anomalies
5.0	System Integration	Integrate market modeling, sentiment analysis, and financial document analysis modules

5.1	API Development	Implement APIs for communication between system co mponents
6.0	Dashboard Development	Design and develop an investor dashboard for trends, anomalies, and explanations
6.1	Data Visualization	Implement charts, anomaly markers, risk panels, and explanation views
7.0	System Testing	Conduct functional testing, anomaly validation, and overall performance evaluation
7.1	Model Evaluation	Evaluate predictive accuracy, anomaly detection quality, interpretability, and consistency over time
8.0	Deployment	Deploy system prototype and prepare demonstration
9.0	Documentation	Prepare final research documentation and reports
10.0	Final Presentation	Prepare project presentation and demonstration

GANTT CHART

Project Activity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Topic selection & supervisor discussions	■										
Proposal preparation	■	■	■								
Proposal submission & presentation			■								
Literature review & system design		■	■	■							
Data collection (CSE data, reports, news)			■	■	■						
Core module development				■	■	■	■				
Progress Presentation 1					■						
System integration & dashboard development					■	■	■	■			
Progress Presentation 2								■			
Testing & performance evaluation							■	■	■		
Final report writing								■	■	■	
Final report submission										■	
Website submission										■	
Final presentation & viva										■	
Research paper submission										■	
Research paper publication evidence											■

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